

## VOTING CLASSIFIER WITH K-FOLD CROSS VALIDATION :

### FOR K =1 CROSS VALIDATION IS NOT POSSIBLE:

Cross-validation with K=1 is not possible because it requires splitting the data into zero testing sets, leaving no data to evaluate the model.

### Why K=1 Fails

- **No Validation Set:** K-fold cross-validation divides data into K equal parts. If K=1, the split is 1 group.
- **Zero Training or Testing Split:** You use the whole dataset to train the model and have nothing left to test it.
- **Math Breakdown:** The testing set size is  $\frac{1}{K}$  of the total data. For K=1, the test size is  $\frac{1}{1} = 1$  (the whole dataset), meaning the training set size becomes  $1 - 1 = 0$ . You cannot train a model with zero data

So we'll take fold 1 when k=2 .

### Fold 1 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      1.0000
Gradient Boosting prob:  0.4744
Logistic Regression prob:0.4759
Average Ensemble prob:   0.6501
Prediction Threshold:    0.64
Final Predicted Class:   1
```

#### Ensemble Voting Breakdown - Fold 1 (Top 5 Test Samples):

| Idx | RF   | GB   | LR   | Average | FinalClass |
|-----|------|------|------|---------|------------|
| 1   | 1.00 | 0.47 | 0.48 | 0.65    | 1          |
| 2   | 1.00 | 0.97 | 0.42 | 0.80    | 1          |
| 3   | 1.00 | 0.88 | 0.60 | 0.83    | 1          |
| 4   | 1.00 | 0.96 | 0.83 | 0.93    | 1          |
| 5   | 0.98 | 0.38 | 0.61 | 0.66    | 1          |

### Fold 2 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      1.0000
Gradient Boosting prob:  0.9526
Logistic Regression prob:0.8918
Average Ensemble prob:   0.9481
Prediction Threshold:    0.64
Final Predicted Class:   1
```

### Fold 3 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      1.0000  
Gradient Boosting prob:  0.9608  
Logistic Regression prob:0.8501  
Average Ensemble prob:   0.9370  
Prediction Threshold:    0.64  
Final Predicted Class:   1
```

### Fold 4 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      0.9900  
Gradient Boosting prob:  0.7989  
Logistic Regression prob:0.5455  
Average Ensemble prob:   0.7781  
Prediction Threshold:    0.64  
Final Predicted Class:   1
```

### Fold 5 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      0.9800  
Gradient Boosting prob:  0.9561  
Logistic Regression prob:0.7897  
Average Ensemble prob:   0.9086  
Prediction Threshold:    0.64  
Final Predicted Class:   1
```

### Fold 6 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      0.9900  
Gradient Boosting prob:  0.8914  
Logistic Regression prob:0.4699  
Average Ensemble prob:   0.7838  
Prediction Threshold:    0.64  
Final Predicted Class:   1
```

### Fold 7 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      0.9900
Gradient Boosting prob:  0.9704
Logistic Regression prob:0.8538
Average Ensemble prob:   0.9381
Prediction Threshold:    0.64
Final Predicted Class:   1
```

### Fold 8 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      0.9900
Gradient Boosting prob:  0.8883
Logistic Regression prob:0.6453
Average Ensemble prob:   0.8412
Prediction Threshold:    0.64
Final Predicted Class:   1
```

### Fold 9 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      0.9700
Gradient Boosting prob:  0.9444
Logistic Regression prob:0.3883
Average Ensemble prob:   0.7676
Prediction Threshold:    0.64
Final Predicted Class:   1
```

### Fold 10 - VotingClassifier Decision for One Sample:

```
Random Forest prob:      1.0000
Gradient Boosting prob:  0.7866
Logistic Regression prob:0.6148
Average Ensemble prob:   0.8005
Prediction Threshold:    0.64
Final Predicted Class:   1
```

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✓ FINAL RESEARCH SUMMARY:

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Final Average Accuracy: 93.64%

Final Average Sensitivity: 89.93%

Final Average Specificity: 97.52%

Final Average Precision: 97.84%

Final Average Recall: 89.93%

Final Average F1 Score: 93.67%

Final Average Kappa Score: 87.19%

Final Average AUC: 98.97%

Here the data set that used is:"

<https://docs.google.com/spreadsheets/d/1GwUew6wtbX0OPm9vGSfs4hSu1spH2wDi0MBiLwcGuIY/edit?usp=sharing> “

The code : “

[https://colab.research.google.com/drive/1F-QNBrrR6JFIECzQ\\_kkn2GrgsR\\_pUvGc?usp=sharing](https://colab.research.google.com/drive/1F-QNBrrR6JFIECzQ_kkn2GrgsR_pUvGc?usp=sharing)  
“  
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## Heart Disease Ensemble Model Performance

| Evaluation Metric | Score (Percentage) |
|-------------------|--------------------|
| Accuracy          | 93.64403917595408  |
| Sensitivity       | 89.93125746129378  |
| Specificity       | 97.51764565641301  |
| Precision         | 97.8385528448046   |
| Recall            | 89.93125746129378  |
| F1 Score          | 93.66685030373272  |
| Kappa Score       | 87.1914752279874   |
| AUC               | 98.96539902903318  |